

Signal Classification based on Spectral Correlation Analysis and SVM in Cognitive Radio

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Abstract

Spectrum sensing is one of the most challenging functions in cognitive radio system. Detection of the presence of signals and distinction of the type of signals in a particular frequency band are critical for cognitive radios to adapt to the radio environment. In this paper, a novel approach to signal classification combining spectral correlation analysis and support vector machine (SVM) is introduced. Four spectral coherence characteristic parameters are chosen via spectral correlation analysis. By utilizing a nonlinear SVM, a significant amount of calculation is performed offline, thus the computational complexity is reduced. Simulations indicate that the overall success rate is above 92.8% with data length of 1000 when SNR is equal to 4 dB. Compared to the existing methods including the classifiers based on binary decision tree (BDT) and multilayer linear perceptron network (MLPN), the proposed approach is more effective in the case of low SNR and limited training numbers.

1. Introduction

Cognitive radio, defined as an intelligent wireless communication system that is aware of its environment and adapts its statistical variations according to learning from it, is viewed as an efficient approach for improving the utilization of the precious spectrum resource [1,2]. Spectrum sensing is an important function which enables the cognitive radio to share the spectrum in licensed bands by detecting temporarily unused spectral resources. Moreover, spectrum sensing can not be restricted to simply monitor the power in some frequency bands of interest but must include detection and identification in order to avoid interference [3,4].

Recent research efforts exploit the cyclostationary features [5]-[7] of signals via spectral correlation

analysis as a method for spectrum sensing, which has been found to be superior to simple energy detection and matched filtering. Energy detection can only detect whether or not a signal is present and utilizing a matched filter system requires extensive knowledge about the channel and signals that are to be identified. The method, which is not susceptible to in-band interference, can be used to detect and classify different types of signal. Conventional signal classification approaches are mainly based on decision theoretic [11] and statistical pattern recognition [6]. In [11], a binary decision tree is designed to signal classification. However, it's hard to obtain the decision thresholds and rules, which needs a large amount of calculation. For more efficient and reliable performance, a novel approach based on multilayer linear perceptron network for signal classification in cognitive radio is proposed in [6].

Support vector machine (SVM) is a new statistical pattern recognition approach, which is based on structural risk minimization principle. Compared with the conventional methods based on empirical risk minimization like artificial neural network (ANN), it has been found to give better generalization and better performance for small training examples. In this paper, a novel approach of signal classification for cognitive radios combining the spectral correlation analysis and SVM is proposed. The high performance and robustness of the method is proved by simulation experiments.

The rest of the paper is organized as follows: Section 2 introduces the spectral correlation analysis of the communication signals. In Section 3, four spectral coherence characteristic parameters are extracted as the input vector of the classifier. In Section 4, we present a nonlinear SVM classifier combining the spectral correlation features. Simulation results are presented in Section 5, and several concluding remarks are made in Section 6.

2. Spectral Correlation Analysis

Many signals used in communication systems exhibit periodicities of their second order statistical parameters due to the operations such as sampling, modulating, multiplexing and coding. These cyclostationary properties, which are named as spectral correlation features, can be used for signal detection and recognition.

In order to analyze the cyclostationary features of the signal $x(t)$, two key functions are typically utilized. The cyclic autocorrelation function (CAF) is used for time domain analysis, which can be expressed as

$$R_x^\alpha(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t + \frac{\tau}{2}) x(t - \frac{\tau}{2}) e^{-j2\pi\alpha t} dt \quad (1)$$

The spectral correlation function (SCF), which exhibits the spectral correlation of the signal $x(t)$, is obtained from the Fourier transform of the cyclic autocorrelation in Equation (1).

$$\begin{aligned} S_x^\alpha(f) &= \int_{-\infty}^{+\infty} R_x^\alpha(\tau) e^{-j2\pi f\tau} d\tau \\ &= \int_{-\infty}^{+\infty} \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t + \frac{\tau}{2}) x(t - \frac{\tau}{2}) \cdot \\ &\quad e^{-j2\pi\alpha t} e^{-j2\pi f\tau} dt d\tau \\ &= \lim_{T \rightarrow +\infty} S_{xT}^\alpha(t, f) \end{aligned} \quad (2)$$

where α is the cycle frequency, f is the spectral frequency and $S_{xT}^\alpha(t, f)$ is the cyclic periodogram of $S_x^\alpha(f)$.

$$S_{xT}^\alpha(t, f) = \left[X_T(t, f + \alpha/2) \cdot X_T^*(t, f - \alpha/2) \right] / T \quad (3)$$

where the Fourier transform of the function $x(u)$ on the bounded time interval $[t - T/2, t + T/2]$ is defined as

$$X_T(t, f) = \int_{t-T/2}^{t+T/2} x(u) e^{-j2\pi fu} du \quad (4)$$

The correlation coefficient for the SCF between frequency components $f \pm \alpha/2$, which is known as spectral coherence coefficient (SCC), can be calculated as

$$C_x^\alpha(f) = \frac{S_x^\alpha(f)}{\sqrt{S_x^0(f + \alpha/2) \cdot S_x^0(f - \alpha/2)}} \quad (5)$$

The magnitude of the SCC ranges from 0 to 1 with $\alpha = 0$ for all f . Different types of signal (i.e. AM, ASK, FSK, PSK, MSK, QPSK) can be distinguished based on several characteristic parameters of SCF and SCC. The method for signal feature extraction

presented in the next section is based on the above analysis.

3. Signal Feature Extraction

In practical situations, however, the number of observation samples at the sensor is limited. Therefore, the spectral correlation function needs to be estimated from a finite set of samples. In general, two methods are used for spectral correlation estimation including time-domain averaging and frequency-domain smoothing. In this paper, the frequency-smoothing method is used for spectral correlation estimation, which can be expressed as follows.

$$\begin{aligned} S_x^\alpha(f)_{\Delta f} &= \frac{1}{M} \sum_{n=(M-1)/2}^{(M+1)/2} \frac{1}{\Delta f} X_{\Delta t}(t, f + \frac{\alpha}{2} + nF_s) \cdot \\ &\quad X_{\Delta t}^*(t, f - \frac{\alpha}{2} + nF_s) \end{aligned} \quad (6)$$

$$X_{\Delta t}(t, f) = \sum_{k=0}^{N-1} W_{\Delta t}(kT_s) \cdot X(t - kT_s) e^{-j2\pi f(t - kT_s)}$$

where $F_s = 1/(N-1)T_s$ is the frequency increment, T_s is the cyclic sampling interval and the length of the sample is $N = \Delta t/T_s$. Thus, the data-tapering window $W_{\Delta t}(kT_s)$ is of the width $\Delta f = M \cdot F_s$.

Based on the calculation of the spectral correlation function, we can obtain the spectral correlation magnitude surface of different types of signal. According to above analysis, several spectral correlation features can be extracted for distinction of different modulation types [9, 10]. Typically, four key features which are sensitive with modulation types and insensitive with SNR variation are listed below:

- Number of δ pulse on f -domain of SCF: x_1

Let $\alpha = 0$ in Equation (6), the SCF is transformed into $S_x^0(f)$, thus x_1 can be obtained from the ichnography of $S_x^0(f)$.

- Number of cyclic spectral line on α -domain of SCF: x_2

Let $f = 0$ in Equation (6), the SCF is transformed into $S_x^\alpha(0)$, and x_2 can be obtained from the ichnography of $S_x^\alpha(f)$.

- Average energy of cyclic spectral line on α -domain of SCF: x_3

The Average energy of cyclic spectral line on α -domain of SCF can be computed by the equation as follows:

$$x_3 = \int_0^{+\infty} |S_x^\alpha(f)|^2 df \quad (7)$$

- Maximum value of SCC: x_4

The spectral coherence coefficient can be obtained via Equation (5) and Equation (6). Then, the maximum value of SCC is the key feature x_4 .

In the experiments, 4000 features are extracted from the signal for every trial. In order to prevent numerical computational errors, the features need to be normalized by subtracting mean of each feature from the original feature and dividing the result by the standard deviation of the same feature.

$$x'_i = (x_i - \bar{x}_i) / \delta_{x_i} \quad i = 1, 2, 3, 4 \quad (8)$$

After normalization, the feature vector $\mathbf{x} = (x'_1, x'_2, x'_3, x'_4)$ is used as the input of the signal classifier.

4. SVM based Signal Classifier

The support vector machine (SVM) was coined by Vapnik in the late 1960s on the foundation of statistical learning theory. It was originally developed for binary classification problem. The optimal solution of SVM for a linearly separable case was introduced by Vapnik. Later this was extended to non-separable cases by [8]. Since the classification of communication signals is obvious to be a linearly non-separable problem, we will only discuss the computation of this optimization problem in this paper.

Given linearly non-separable classification problem, we suppose a training group is $\{(\mathbf{x}_i, y_i)\}$, $i = 1, \dots, l$, where $\mathbf{x} \in R^N$, $y \in \{+1, -1\}$. Assumed that the group can be classified by a hyperplane which is defined as

$$\mathbf{w}^T \Phi(\mathbf{x}) + b = 0 \quad (9)$$

where the vector $\mathbf{w}$ defines the boundary of different classes of data, b is a scalar threshold and $\Phi(\mathbf{x}) = \|\mathbf{x}\|^2 / 2$.

In order to solve this non-separable problem, the non-minus slack variables ξ_i are introduced, and then the training vectors must satisfy

$$y_i(\mathbf{w}^T \mathbf{x}_i + b_i) - 1 \geq 1 - \xi_i, \quad i = 1, 2, \dots, l \quad (10)$$

The hyperplane, which makes $\Phi(\mathbf{w}) = \|\mathbf{w}\|^2 / 2$ to be minimum, is named as the optimal hyperplane. All of the training vectors are correctly classified by it and the vectors of each class are separated with a maximum margin.

Usually, structuring a hyperplane is solved as a quadratic optimization problem that can be formulated as

$$\begin{aligned} \min_{\mathbf{w}, b, \xi_i} \quad & \frac{1}{2} \mathbf{w}^T \mathbf{w} + C \sum_{i=1}^l \xi_i \\ \text{s. t.} \quad & y_i(\mathbf{w}^T \Phi(\mathbf{x}_i) + b) \geq 1 - \xi_i \\ & \xi_i \geq 0, \quad i = 1, 2, \dots, l \end{aligned} \quad (11)$$

where C is the penalty parameter, which is used to control the training error rate by different values.

Using a Lagrange multiplier technique, the optimization problem can be converted into

$$\begin{aligned} F = \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^l \xi_i - \sum_{i=1}^l \alpha_i [y_i(\mathbf{w}^T \Phi(\mathbf{x}_i) + b) - 1 + \xi_i] \\ - \sum_{i=1}^l \beta_i \xi_i \end{aligned} \quad (12)$$

where $\alpha_i, \beta_i \geq 0$ are Lagrange multiplier factors. By computing the derivatives with respect to $\mathbf{w}, b, \xi_i$, the dual representation of the optimization problem can be obtained as

$$\begin{aligned} \max_{\alpha} \quad & \sum_{i=1}^l \alpha_i - \frac{1}{2} \sum_{i,j=1}^l \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j) \\ \text{s. t.} \quad & 0 \leq \alpha_i \leq C, i = 1, 2, \dots, l \\ & y^T \alpha = 0 \end{aligned} \quad (13)$$

where $K(\mathbf{x}_i, \mathbf{x}_j) = \Phi(\mathbf{x}_i) \Phi(\mathbf{x}_j)$ is the kernel function, which is utilized for mapping the input data to higher dimensional space in order to reduce the computational load.

The resulting decision function is obtained as follows:

$$f(x) = \text{sgn} \left\{ \sum_{i=1}^l \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b \right\} \quad (14)$$

There are different kernel functions like polynomial, sigmoid and radial basis function (RBF) used in SVM. A proper kernel functions for a certain problem is dependent on the specific data and till now there is no good method on how to choose a kernel functions. In this paper, the choice of the kernel functions is studied empirically and optimal results are achieved using RBF kernel function.

$$K(\mathbf{x}, \mathbf{x}_i) = \exp \{ -\|\mathbf{x} - \mathbf{x}_i\|^2 / \sigma^2 \} \quad (15)$$

The width of the RBF kernel parameter σ can be determined in general by an iterative process selecting an optimum value based on the full feature set.

The architecture of the SVM classifier combining spectral correlation analysis is shown in Figure 1.

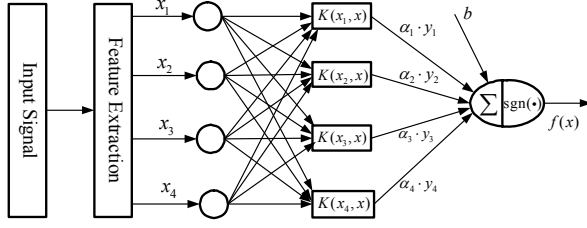


Figure 1. Architecture of the SVM classifier combining spectral correlation analysis.

5. Simulation Results

In this section, a variety of Monte Carlo simulations are presented to illustrate the performance of the algorithm. We compare the proposed SVM classifier to two existing classifiers in this paper. For the simulations, we define a signal set as {AM, ASK, FSK, PSK, MSK, QPSK}. To distinguish 6 modulation types, simulations are carried out with 1024 samples at SNR ranging from 0 dB to 20 dB. In order to simulate the wireless environment of cognitive radios, the channel is assumed to have a multi-path (4-path) delay profile with 150ns delay spread.

Table 1 indicates the probability of correct classification (P_{cc}) for each modulation type with the training data length of 1000. Results show the overall correct rate is above 92.8% for a SNR of 4dB, and 97.3% for a SNR of 8dB. These good results for signals with low SNR in the cognitive radio environment show the proposed approach is insensitive with SNR variation, which come from the effects of the robustness of SVM classifier.

Signal Type	SNR(dB)					
	0dB	4dB	8dB	12dB	16dB	20dB
AM	96.9	100	100	100	100	100
ASK	95.2	100	100	100	100	100
FSK	92.1	97.2	100	100	100	100
PSK	86.3	93.0	98.2	99.9	100	100
MSK	88.6	95.5	99.3	100	100	100
QPSK	85.1	92.8	97.3	99.7	99.9	100

Table 1: Probability of correct classification of SVM classifier (%).

Figure 2 shows the performance of SVM classifier with data length as parameter. When the data length is

100 and for a SNR of 4dB, the P_{cc} is up to 80.6% and with data length 200 and for a SNR of 6dB, the P_{cc} is increased to 90%. When the data length is 1000 and for a SNR of 10dB, the P_{cc} is close to 100%. Above results show that the performance of the SVM classifier is high for small training data.

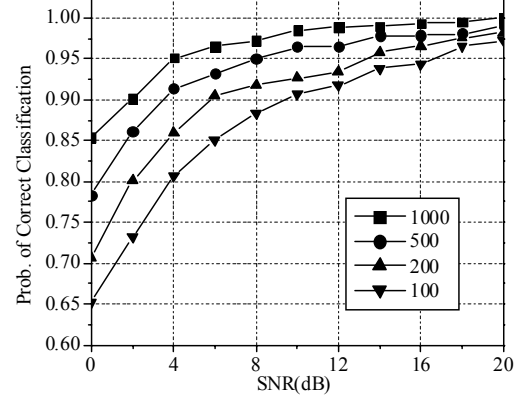


Figure 2: The performance of SVM classifier with different data length.

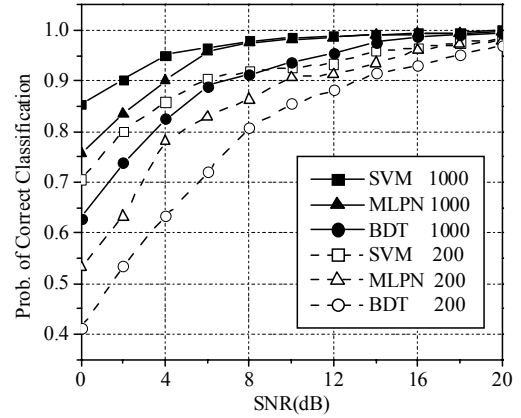


Figure 3. Comparison between three classifiers with different data length.

Figure 3 is the performance comparison between binary decision tree (BDT) based classifier [11], multilayer linear perceptron network (MLPN) classifier [6] and SVM classifier with the spectral correlation features, which is calculated for different signals using data length of 200 and 1000, respectively. It is shown that when the SNR is lower, the multilayer linear perceptron network classifier shows poor performance. While the SNR is higher, the probability of correct classification is increased. In lower SNR, the variation

of the spectral correlation features (SCFs) is drastically due to the effect of the noise. Thus the construction of neural network is not complete with small training data, which results in the performance degradation. The decision tree based classifier only use the partial information of the spectral correlation features (SCFs), therefore, the correct probability is lower than SVM classifier in the whole SNR range. All the results show the high performance of SVM classifier based on spectral correlation features (SCFs).

6. Conclusion

In this paper, we proposed a novel approach combining the spectral correlation features and SVM for signal classification in cognitive radio environment. Four spectral correlation characteristic parameters were selected as feature vector of SVM classifier. Simulation results show that the overall success rate is above 92.8% with data length of 1000 when SNR is equal to 4dB. Compared to the methods in [6] and [11], the proposed approach is more effective in the case of low SNR and limited training numbers.

Future work in the area of signal classification for cognitive radio systems will involve the analysis of higher order spectrum correlation features of more communication signals. Based on these features, a multi-class SVM classifier can be used to improve the accuracy of classification and reduces the computational complexity. In addition, the classifier performance will be examined via simulations using several different channel models.

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